Powertrain Control System

High-Level Design (HLD)

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This document describes the software architecture of the powertrain control domain: engine management, transmission control, torque coordination and emissions aftertreatment components and interfaces.

1 Revision history

Revision	Date	Author	Description
D	2026-05-14	A. Kumar	Initial architecture baseline

2 Scope and context

2.1 Scope

The powertrain domain covers engine torque production, gear ratio management, torque limitation and aftertreatment monitoring. Out of scope: combustion calibration maps, hardware sensor pin mapping and production software implementation details.

2.2 System context

The domain runs on PowertrainEcu. Torque requests from the vehicle motion and ADAS domains arrive via PowertrainGateway over the chassis CAN bus at 500 kbit/s.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
EngineManagementUnit	ApplicationSwComponentType	Engine torque and fuelling control
TransmissionController	ApplicationSwComponentType	Gear ratio and shift control
TorqueCoordinator	ApplicationSwComponentType	Cross-domain torque arbitration
AftertreatmentMonitor	ApplicationSwComponentType	Emissions aftertreatment supervisi
StartStopManager	ApplicationSwComponentType	Start-stop strategy and restart lo
PowertrainSignalService	ServiceSwComponentType	Powertrain signal gateway service
PowertrainGateway	EcuAbstractionSwComponentType	Cross-domain gateway exchange
PowertrainMainComposition	CompositionSwComponentType	Composition of the domain componen

3.2 Ports and interfaces

Port	Direction	Interface	Component
TorqueRequestIn	required	TorqueRequestI	TorqueCoordinator
EngineTorqueOut	provided	EngineTorqueI	EngineManagementUnit
GearRatioOut	provided	GearRatioI	TransmissionController
AftertreatmentStatusOut	provided	AftertreatmentStatusI	AftertreatmentMonitor
StartStopStateOut	provided	StartStopStateI	StartStopManager
PowertrainGatewayIn	required	PowertrainSignalSetI	PowertrainSignalService
GatewayToMotion	provided	VehicleSpeedI	PowertrainGateway
GatewayToAdas	provided	MotionCommandI	PowertrainGateway

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
RequestedTorque	TorqueRequestI	TorqueCoordinator	EngineManagementUnit	12
ActualEngineTorque	EngineTorqueI	EngineManagementUnit	TorqueCoordinator	12
CurrentGearRatio	GearRatioI	TransmissionController	TorqueCoordinator	8
CatalystTemperature	AftertreatmentStatusI	AftertreatmentMonitor	EngineManagementUnit	12
StartStopActive	StartStopStateI	StartStopManager	TransmissionControlle	1
EngineSpeed	EngineSpeedI	EngineManagementUnit	TransmissionControlle	16
TransmissionMode	TransmissionModeI	TransmissionController	StartStopManager	4

4 Functional flows

4.1 Torque arbitration flow

TorqueCoordinator consumes TorqueRequestIn from the vehicle motion domain and MotionCommandI from the ADAS domain, applies priority limits and publishes RequestedTorque to EngineManagementUnit. EngineManagementUnit confirms through ActualEngineTorque.

4.2 Shift intervention flow

TransmissionController consumes EngineSpeed and CurrentGearRatio to request temporary torque reduction from TorqueCoordinator during shift events. StartStopManager suspends start-stop during active shifts.

4.3 Aftertreatment flow

AftertreatmentMonitor tracks CatalystTemperature and raises a limp-home request to EngineManagementUnit when thresholds are exceeded. DiagnosticsFrontend reads the fault state over DiagnosticServiceI.

5 Timing and runnables

TimingEvent TorqueArbitrationEvent triggers CoordinateTorqueRunnable every 10 ms. TimingEvent ShiftMonitorEvent triggers MonitorShiftRunnable every 20 ms. TimingEvent AftertreatmentSampleEvent triggers SampleAftertreatmentRunnable every 100 ms. All runnables are scheduled by PowertrainTask on PowertrainEcu.

6 Basic software integration

Signal routing uses Com and PduR. Incoming frames are handled by CanIf on PowertrainEcu. NvM stores the shift strategy calibration block. Rte provides the port API used by all powertrain application components.

Mode management is performed by BswM and EcuM. Watchdog supervision for PowertrainTask is performed by WdgM. The gateway maps IPdu to NPdu inside PowertrainGateway before transmission.

The diagnostic path routes UDS requests through Dcm and Dem. PowertrainSignalService consumes PowertrainSignalSetI and forwards routine results to the tester.

7 Diagnostics and configuration

DTCs for catalyst overtemperature, shift timeout and torque deviation are defined in the diagnostics specification. The HLD only defines the reporting path: component to Dcm to CanIf.

Configuration parameters are grouped into the domain configuration container and written through NvM during shutdown.

8 Known limitations

TorqueCoordinator priority arbitration between ADAS and motion requests is fixed at compile time in this revision.

9 Open questions

Open: calibration ownership for cross-domain interfaces and the timeline for production software freeze are tracked outside this document.